The differential labelling thus enabled them to identify DNA and proteins of the phage.

Hershey and Chalse mixed the labelled phages with unlabeled *E. coli* and allowed bacteriophages to attack and inject their genetic material. Soon after infection (before lysis of bacteria), the bacterial cells were gently agitated in a blender to loosen the adhering phase particles. It was observed that only ^{32}P was found associated with bacterial cells and ^{35}S was in the surrounding medium and not in the bacterial cells. When phage progeny was studied for radioactivity, it was found that it carried only ^{32}P and not ^{35}S **(Fig. 5.2)**. These results clearly indicate that only DNA and not protein coat entered the bacterial cells. Hershey and Chase thus conclusively proved that it was DNA, not protein, which carries the hereditary information from virus to bacteria.

5.4 Chemistry of Nucleic Acids

Having identified the genetic material as the nucleic acid DNA (or RNA), we proceed to examine the chemical structure of these molecules. Generally nucleic acids are a long chain or polymer of repeating subunits called nucleotides. Each nucleotide subunit is composed of three parts: a nitrogenous base, a five carbon sugar (pentose) and a phosphate group.

Pentose sugar

There are two types of nucleic acids depending on the type of pentose sugar. Those containing deoxyribose sugar are called **Deoxyribo Nucleic Acid** (DNA) and those with ribose sugar are known as **Ribonucleic Acid** (RNA). DNA is found in the nucleus of eukaryotes and nucleoid of prokaryotes. The only difference between these two sugars is that there is one oxygen atom less in deoxyribose.

Nitrogenous bases

The bases are nitrogen containing molecules having the chemical properties of a base (a substance that accepts H^+ ion or proton in solution). DNA and RNA both have four bases (two purines and two pyrimidines) in their nucleotide chain. Two of the bases, Adenine (A) and Guanine (G) have double carbon–nitrogen ring structures and are called purines. The bases, Thymine (T), Cytosine (C) and Uracil (U) have single ring structure and these are called pyrimidines. Thymine is unique for DNA, while Uracil is unique for RNA.

The phosphate functional group

It is derived from phosphoric acid (H_3PO_4), has three active OH^- groups of which two are involved in strand formation. The phosphate functional group (PO_4) gives DNA and RNA the property of an acid (a substance that releases an H^+ ion or proton in solution) at physiological pH, hence the name **nucleic acid**. The bonds that are formed from phosphates are esters. The oxygen atom of the phosphate group is negatively charged after the formation of the phosphodiester bonds. This negatively charged phosphate ensures the retention of nucleic acid within the cell or nuclear membrane.

Nucleoside and nucleotide

The nitrogenous base is chemically linked to one molecule of sugar (at the 1-carbon of the sugar) forming a **nucleoside**. When a phosphate group is attached to the 5' carbon of the same sugar, the nucleoside becomes a **nucleotide**. The nucleotides are joined (polymerized) by condensation reaction to form a polynucleotide chain. The hydroxyl group on the 3' carbon of a sugar of one nucleotide forms an ester with the phosphate of another nucleotide. The chemical bonds that link the sugar

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